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Assignment – II

Database Management System
(24CSH-298)

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Submitted To:

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Question 1: Explain Each type of Security Issues.

Database security issues refer to risk or problems that may lead to unauthorized access, misuse, or loss of data. These issues can be classified into following types:

(1) Legal and ethical issues

Legal and ethical concern that laws, regulations, and moral responsibilities related to the storage and use of data in a database system.

Organizations must ensure that they comply with data protection laws and privacy regulations.

Sensitive information such as personal data, financial records and confidential business information should not be accessed or shared without proper authorization.

Example: Leaking a customer personal information or using employee data without consent.

(2) Policy Issues

Policy issues relate to the rules and guidelines established by an organization to protect its database systems. Organizations develop security policies that defines how data should be accessed, used and protected.

These policies specify user roles, access rights, responsibility of employees when handling data.

3) System Related Issue

System Related issues arise from technical weakness in database system or infrastructure. like weak authentication systems, software vulnerabilities, Improper configuration of database servers and lack of updates or security patches

Example: A hacker exploiting a database vulnerability to steal information

Question 2: Explain the different database security control techniques

Answer: Database security control techniques are methods used to protect databases from unauthorized access and misuse

(i) Access Control

Access Control restricts who can access the database and what operations they can perform, Users are given specific permission (read, write, update, delete). Implemented using user accounts, roles and privileges. Ensures that only authorized users can access sensitive data.
eg. A clerk can view records but cannot modify them

(ii) Inference Control

Inference control prevents users from deducing sensitive information from accessible data.

eg: Preventing someone from calculating an individual's average salary queries

(iii) Flow Control

Flow control ensures that information does not flow from higher security levels to lower security levels.

It restricts how information moves within a system
example: preventing secret data from being transferred to public databases

(iv) Data Encryption

Data encryption converts plain text data into coded form cipher text to protect it from unauthorized access

example: encrypting credit card details stored in a database.

Question: Consider a table Employee(EmpID, EmpName, Salary, DeptID). Write an SQL query to create a view that displays only EmpID, EmpName, and DeptID from the Employee table.

Answer: A view is a virtual table created from one or more tables. It stores a query rather than actual data

SQL Query:

```
CREATE VIEW Employee-View AS  
SELECT EmpID, EmpName, DeptID  
FROM Employee;
```

This view will display only EmpID, EmpName, and DeptID, hiding other columns such as Salary